Coherence Collapse: Diagnosing Why Code Agents Fail After Reaching the Right Code

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PROBLEM

Code agents resolve **65–70%** of SWE-bench issues, but **Pass@1** cannot tell us *why* the rest fail — and capable-model failures are misdiagnosed without trajectory data.

GAP

Outcome-only metrics collapse a trajectory into one bit — hiding the real cause of failure.

MOTIVATION

- **Every task ships a reference patch** — the exact files & functions that must change.
- It turns messy tool-use logs into checkable **localization, comprehension, and modification** signals at every stage.

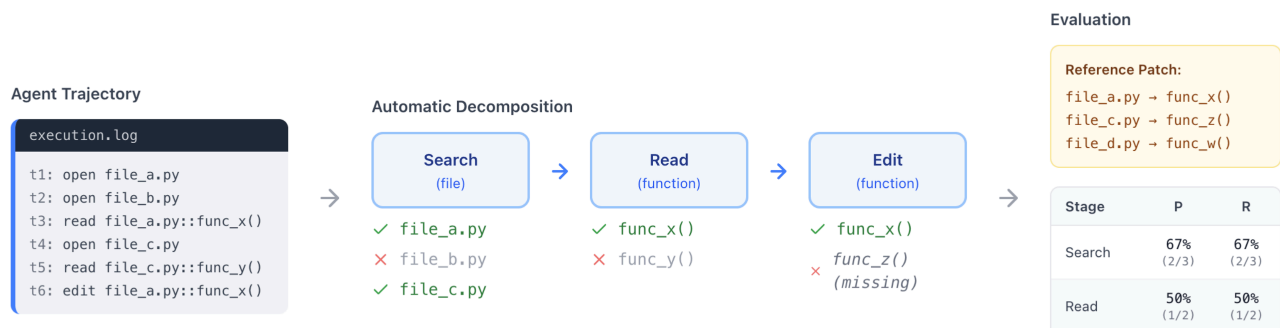
> *Without trajectory data, a failure that reached the right code looks identical to one that never found it.*

TAKEAWAY

For capable code agents the frontier bottleneck is no longer *finding* the right code — it is **not destroying the fix once found**. In 5 EXACT cases the agent produced a patch **bit-identical to the gold reference** mid-trajectory, then overwrote it — an edit-commit checkpoint recovers all 5. A reference-free consensus variant even lifts GPT-5 Pass@1 by a directional **+3.0 points** without any reference patch.

Coherence Collapse is a distinct, recoverable failure mode — trajectory-level analysis is what makes it visible.

METHOD



TrajEval decomposes each agent execution trace into reference-patch-aligned Search (file), Read (function), and Edit (function) stages, then scores stage-wise precision and recall against the gold patch.

1 Three-Stage Decomposition

training-free

Each trace is split into **Search** (file-level), **Read** (function-level inspection), and **Edit** (function-level modification), each compared against the gold reference set of files and functions.

$$P_s = \frac{|A_s \cap G_s|}{|A_s|}, \quad R_s = \frac{|A_s \cap G_s|}{|G_s|}$$

precision & recall per stage $s \in \{\text{search, read, edit}\}$; A_s = agent set, G_s = gold set

2 Stage-Wise Precision & Recall

per trace

Trace → Search → Read → Edit → P / R

Each stage yields **precision** and **recall** vs. the reference patch — turning one binary Pass@1 into a six-number profile that pinpoints *which* stage broke. Applied across **3 architectures** (SWE-Agent, OpenHands, bash-only LiveSWEAgent) and **7 models** from GPT-5 to the Qwen3 family. High **recall** with low **precision** flags an agent that found the target but thrashed around it; the reverse flags an agent that never reached it.

16,758 agent trajectories scored end-to-end

3 Isolating the Edit-Quality Residual

the culprit

The Edit-Quality bucket — **56.9%** of all SWE-Agent failures — is the dominant residual for every capable model, yet stage-level precision and recall alone cannot decompose it into the actionable, mutually-exclusive failure types below.

Right location, wrong code: the failure lives inside the edit itself, not the search.

4 Failure Taxonomy

5 themes / 9 sub-types

All 914 Edit-Quality failures are classified by a cascading-precedence rule over three trajectory signals into these themes:

Coherence Collapse · Confused Thrashing · Near-Correct Corrupted · Semantic Error · Wrong Branch Point · Mis-localization · Scope / Completeness · Execution Gap

EDIT-QUALITY THEME	N	%
Coherence Collapse	**363**	**39.7**
Semantic Error	302	33.0
Intra-File Mis-loc.	161	17.6
Scope / Completeness	72	7.9
Execution Gap	16	1.8

> *Second-author agreement on the 3-class split is substantial (κ=0.80, 90% raw).*

SCAN TO READ

Paper

KEY RESULTS

FAILURE LOCUS	SHARE
Localization (Search + Read)	minority
Edit-Quality (right code, wrong patch)	60–69%

Coherence Collapse = 39.7% of Edit-Quality failures — the largest theme.

SO WHAT → *Most failures **reach and edit the correct functions** yet emit a wrong patch — replicating on PolyBench (32.3%).*

HEADLINE NUMBERS

60–69% **of capable-model failures reach & edit the right code — yet ship a wrong patch**

Failure comes after localization, not before

16,758 trajectorie analyzed

39.7% Coherence Collapse share

5/5 gold patches recovered